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Mechanical property prediction of random copolymers using uncertainty-based active learning

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ABSTRACT

The copolymer, a widely used material in our daily lives, presents a significant challenge in targeted sequence design. While recent advancements in computational simulation and data science offer a promising avenue for addressing this complex issue, challenges persist in labeled data scarcity. In this study, we introduce an uncertainty-based active learning framework for predicting the properties of random copolymers. We found that the active learning strategy allowed for labeling only 40 data points within the design space of 1550 data points, drastically reducing the labeling efforts by 97%. Most data selected by active learning were positioned on the design space's periphery, transforming the learning task into an interpolation problem. Through integrating active learning and molecular dynamics, we successfully overcame the combinatorial explosion problem in copolymer sequence design, streamlining the data labeling process and culminating in a highly accurate model. This research demonstrates data science's potential in polymer design, especially when facing data scarcity.

1. Introduction

Copolymers are polymers composed of two distinct types of monomers. Their unique characteristic lies in combining the advantages of both monomers, providing flexibility to optimize various properties such as dielectric properties [1], thermal responses [2], mechanical properties [3], and more. One common type of copolymer is the hard-soft copolymer, such as styrene-butadiene rubber (SBR) [4], and polyolefin triblock copolymer [5]. These polymers typically consist of hard and soft segments, which form phase-separated structures, enhancing their mechanical properties. Moreover, numerous research reports emphasize that optimizing the properties of copolymers requires careful consideration of the sequence's impact [6–9]. However, achieving sequence control becomes challenging when designing copolymers due to the combinatorial explosion problem. Even with the simplest A-B copolymer, gaining brute-force experimental insights for every possible sequence is impractical.

Machine learning and other data science methods are extensively employed in materials design [10–18]. Due to the scarcity of experimental data, much of such research relies on simulation data, such as Molecular Dynamics (MD) or first-principles theory. These methods provide a novel approach for optimizing copolymer properties compared to traditional trial-and-error approaches. Moreover, more studies

are investing in designing copolymer sequences to achieve optimized properties by integrating machine learning and computational simulations. This innovative combination holds great promise for advancing the field of copolymer design. Examples include employing MD simulation and unsupervised learning to characterize aggregates formed by sequence-defined copolymers [19], combining MD simulation and genetic algorithm to minimize the interface energy between sequence-specific copolymers [20], or using machine learning and simulation data to conduct sequence design for copolymers to optimize the radius of gyration (R_g) [21].

While computational simulation methods contribute significantly to constructing copolymer datasets, certain constraints remain. For instance, when dealing with a random copolymer design problem, the process can be time-consuming and demand substantial computational resources if we aim to label all candidates through brute-force methods. Consequently, there is a need to find efficient methods for data labeling. Active learning is a novel way to mitigate costs associated with labeling data, providing a more resource-effective approach [16,22–24]. The core concept of the active learning process is to obtain feedback from the model. Once we have received feedback, we can make informed decisions on the next steps based on the acquired information. For

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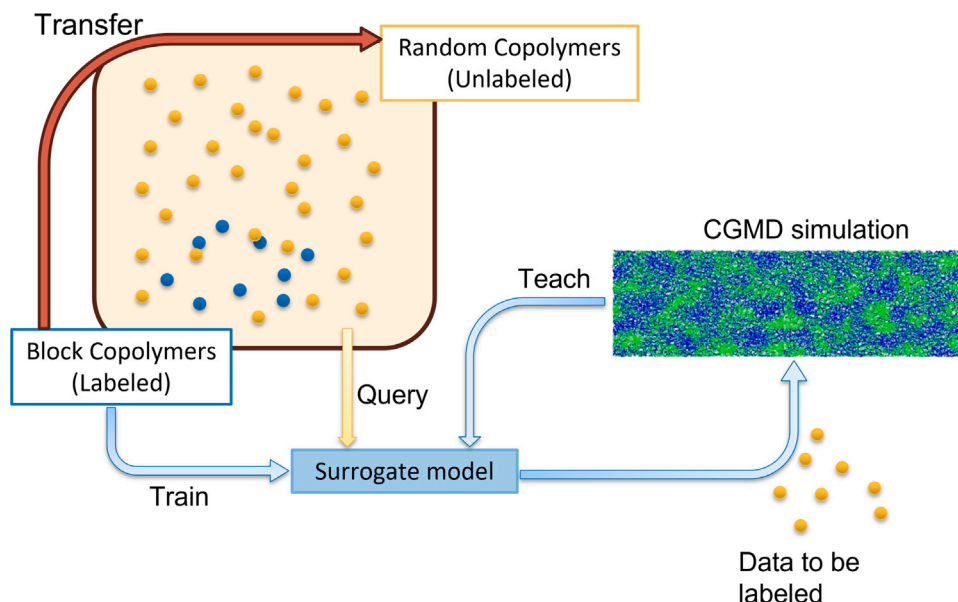


Fig. 1. Research overview: We employed active learning to sample data efficiently and constructed a surrogate model for predicting the mechanical properties of copolymers.

example, we can assess the uncertainty of unlabeled data based on the model feedback and use this information to decide which data to label [23,24]. These methods commonly employ the model's responses to guide data sampling, bringing it closer to the optimized objective and thereby improving the model [25,26]. In copolymer sequence prediction and design, certain studies utilize active learning to aid in constructing a surrogate model and designing an optimal sequence through efficient sampling and labeling. Examples include employing active learning to design dispersants with multiple objectives and combining it with MD simulation and Pareto frontier analysis [27]. Another application uses MD-based active learning to predict and optimize the radius of gyration (R_g) for copolymer sequences [28].

In his seminal work, Aoyagi employed coarse-grained molecular dynamics (CGMD), machine learning, and Bayesian optimization to design thermoplastic elastomers [15]. He constructed a dataset of 1200 ABA-triblock copolymers corresponding to their stress-strain curve (S-S curve), where A and B represent hard and soft segments, respectively. Building on his work, our study centered around transferring the surrogate model's knowledge based on ABA-triblock copolymer to random copolymer design. Given the inherent combinatorial explosion challenge associated with random copolymers, we employed active learning to facilitate the identification of valuable data for labeling within the design space. The labeling process was carried out using CGMD simulation. This study adopts Gaussian Process Regression (GPR) [29–32] as the surrogate model. GPR is selected due to its ability to offer estimates of uncertainty and to develop a predictive model that is both smooth and nonlinear. Additionally, an increasing body of research employs GPR to aid in designing molecules or materials [33].

2. Methods

The research begins by constructing an initial model using the ABA-triblock copolymer dataset from Aoyagi [15]. Once a surrogate model is established, leveraging the foundational knowledge from copolymers and stress-strain curves, random copolymers are tested on the model, and the one with the highest uncertainty is selected. This data is labeled using CGMD simulation, and the surrogate model is updated accordingly. The process iterates to refine the model further, as shown in Fig. 1. The methods used in the process are described below.

2.1. Hard-soft copolymer

This research focuses on the hard-soft copolymer, with SBR as a prime example. SBR, a widely used rubber, comprises two monomers: styrene and butadiene. Styrene is the hard segment, while butadiene is the soft segment. These copolymers are all random, meaning there are no constraints on the distribution of their monomers. The polymer structure is illustrated in Fig. 2. Below, we will use “A” to denote the styrene monomer and “B” to denote the butadiene monomer.

2.2. Sample strategy of random copolymer

In our study, we constrain the chain length of copolymers to be from the range of 60 to 120. All copolymers consist of two monomers, denoted by A and B, representing the hard and soft segments, respectively. Also, we need to consider that the reverse sequence is equal to the forward sequence. Then, we can calculate the number of candidates in the design space:

$$\frac{1}{2} \sum_{i=60}^{120} 2^i \approx 1.33 \times 10^{36} \quad (1)$$

Enumerating all possible sequences in such a tremendous design space would entail significant storage requirements, not to mention the challenges of labeling data through MD or other methods. Therefore, we will employ sampling strategies to facilitate uniform data sampling within this vast design space.

We prioritize maintaining a consistent length to ensure uniformity in the sampled polymers, recognizing the critical role of chain length in determining polymer properties. We control polymers with lengths ranging from 60 to 120, with increments of 2. We sample 50 polymers for each length, totaling $31 \times 50 = 1550$ polymers within our design space. Subsequently, for polymers with the same length, we ensure uniform monomer fractions, recognizing the significance of this factor in copolymer design. Starting with a fraction of A at 0.1, we increase by 0.015 for each subsequent polymer. The polymer sequences are then shuffled to generate our random copolymer design dataset.

This dataset now comprises 1550 random copolymer sequences, each lacking labeled properties. We aim to establish a surrogate model that matches sequences and properties' relationships.

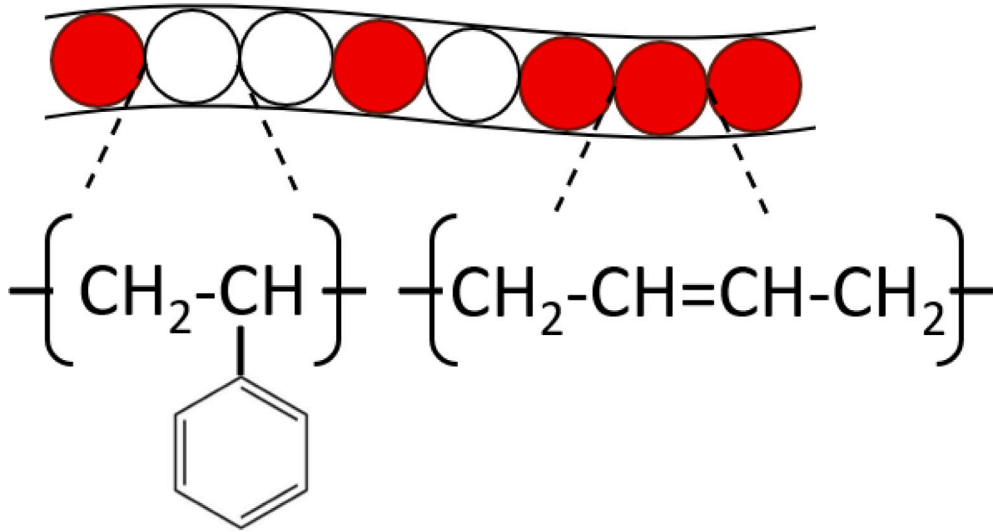


Fig. 2. The styrene-butadiene rubber(SBR). The white beads represent the styrene monomer units, and the red beads represent the butadiene monomer units.

2.3. Simulation

The simulations were conducted by the soft materials simulation system Octa [34,35] and can be divided into the following three steps:

1. Firstly, obtaining the phase-separated structure of copolymers is crucial. Hard-soft copolymers inherently develop microdomain structures [36], and establishing a reliable mechanical property necessitates a well-defined initial phase-separated mesostructure. In this study, we employed self-consistent field theory (SCFT) simulation to acquire the phase-separated structure, and the simulation was conducted using the SUSHI program [37].
2. We used the node density-biased Monte Carlo (NDBMC) method to transform the SCFT phase-separation results into an equivalent bead model.
3. The final step involved utilizing CGMD to simulate the tensile test. Commencing with the obtained phase-separated structure, we utilized the COGNAC [38] program for simulation, ultimately getting the stress-strain (S-S) curve.

The snapshots of the three steps are illustrated in Fig. 3. Additional simulation details can be found in Aoyagi's work [15], encompassing simulation parameters, conditions, and others. The only modification made was to the box size. Given the iterative nature of the active learning process and the aim to swiftly obtain simulation feedback as data labels, the box size was set to $16 \times 16 \times 16$. One simulation process took approximately 5 to 8 h, utilizing the Intel I9-13900k with a 16-core CPU.

2.4. Surrogate model

2.4.1. Model input and output

We defined the model input as seven parameters used to describe a random copolymer. These parameters are listed in Table 1, and an example illustrating the relationship between a sequence and its corresponding model input is shown in Fig. 4. Most of these parameters were designed to capture the clustering of monomers, forming subdomains that significantly influence the mechanical properties.

The model's output was the S-S curve, which we discretized into 101 points. In other words, we divided the S-S curves into 101 discrete points, ranging from strain 0 to 3. This allowed the model to learn and predict based on the corresponding stress values for each discretized strain point.

Table 1

Model input parameters and description.

Parameters	Description
n	number of monomers (degree of polymerization)
n_A	number of A monomers
n_B	number of B monomers
$mean_A$	mean cluster size of A monomers
$mean_B$	mean cluster size of B monomers
max_A	max cluster size of A monomers
max_B	max cluster size of B monomers

2.4.2. Gaussian process regression

The surrogate model employed in this study is Gaussian Process Regression (GPR) [29,39]. GPR is represented as a function combining both mean and covariance, expressed as:

$$f(x) \sim GP(m, k) \quad (2)$$

where $f(x)$ is the real process, GP is the Gaussian process, m denotes the mean function $m(x)$, and k represents the covariance function $k(x_i, x_j)$. Our study defines the mean function using the seven polymer-described parameters, while a kernel object defines the covariance function. The choice of a kernel function is crucial when we train a GPR, as it encapsulates the underlying relationship between input parameters within the covariance matrix. After testing several kernel functions on the dataset, we selected the Matern Kernel, which is defined as:

$$k(x_i, x_j) = \frac{1}{\Gamma(\nu)2^{\nu-1}} \left(\frac{\sqrt{2\nu}}{l} d(x_i, x_j) \right)^\nu K_\nu \left(\frac{\sqrt{2\nu}}{l} d(x_i, x_j) \right) \quad (3)$$

where $d(\cdot, \cdot)$ is the Euclidean distance, K_ν is a modified Bessel function, $\Gamma(\cdot)$ is the gamma function, l is the length scale of the kernel and ν is the parameters which controls the smoothness of the entire function [29]. In our study, ν is set to 1.5, and l would be optimized during training. Also, to avoid the influence of noise, we added the White Kernel to our model. The model is built by Scikit-learn 1.3.0 [40,41].

The learning and prediction process of GPR is grounded in Bayes' theorem concerning prior and posterior probabilities. Upon acquiring a dataset, these data can be utilized to optimize the parameters of our mean and covariance functions through common maximum log-likelihood. With the trained model, we can predict unseen data that are related to the training data. This prediction is facilitated through the joint distribution. Using the most common notation, X is the training

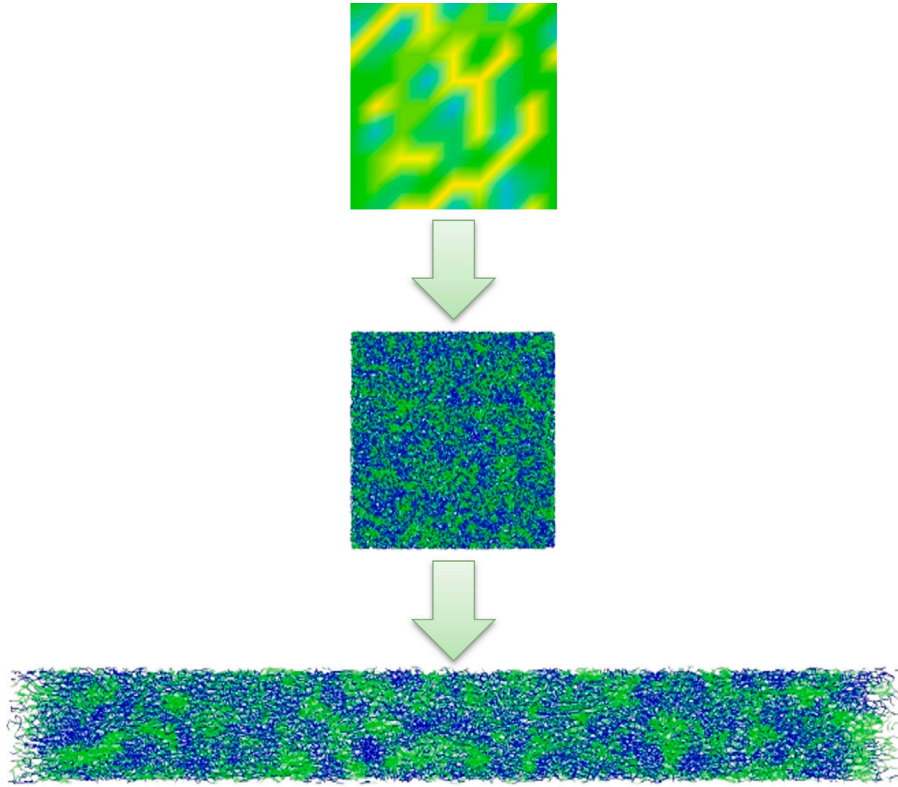


Fig. 3. Snapshot of the 3-step simulation. The first figure is the result of the SCFT calculation, and then we can obtain the bead model's initial state via the NDBMC method, which is the second shot. The third shot is the state after CGMD tensile simulation.

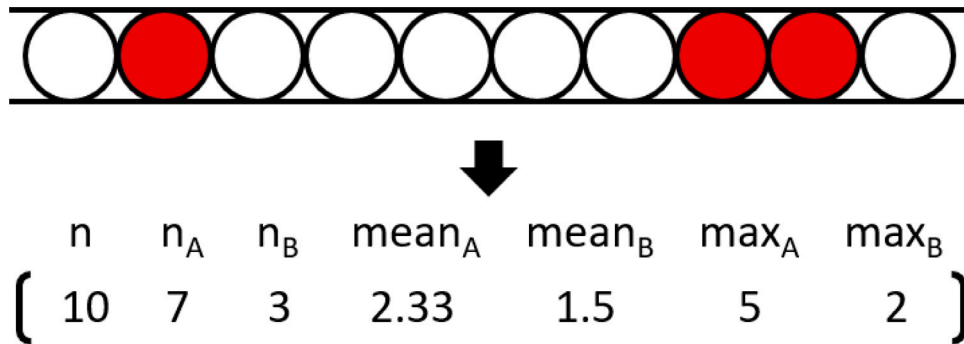


Fig. 4. Illustration of the relationship between a sequence and its corresponding model input.

set input with the known label f , and X^* denotes the unknown dataset to be predicted. The joint distribution is expressed as:

$$\begin{bmatrix} f \\ f_* \end{bmatrix} \sim N \left(\begin{bmatrix} m(X) \\ m(X_*) \end{bmatrix}, \begin{bmatrix} K & K_* \\ K_*^T & K_{**} \end{bmatrix} \right) \quad (4)$$

where f is the joint distribution of the training data, f_* is the joint distribution of the testing data, $K = K(X, X)$, $K_* = K(X, X_*)$ and $K_{**} = K(X_*, X_*)$. So, we can obtain the posterior distribution of f_* via the joint distribution obtained from training data.

2.5. Active learning

Obtaining enough labeled data is challenging due to the associated labeling costs, including applying machine learning in polymer science. Consequently, the critical task becomes selecting data with the utmost value for labeling. Active learning is the methodology we employ to address this issue.

The core concept of active learning revolves around employing a query algorithm to enable the model to seek information. We gain

insights into which data points to label through the model-initiated queries, ultimately facilitating significant improvements in the model's performance. Given the diverse conditions and applications, various query strategies are employed in active learning, including uncertainty sampling, expected model change, query by committee, and more [42]. In our research, the query strategy is uncertainty sampling, which directly obtains uncertainty from the GPR surrogate model. The standard active learning workflow is illustrated in Fig. 5, and the active learning process is conducted via the modAL package in this research [43,44].

3. Results and discussion

3.1. Data distribution of copolymers

Our copolymer dataset encompasses random copolymers and block copolymers. All have a length ranging from 60 to 120. The data for block copolymers is sourced from Aoyagi [15], comprising 1200 ABA-triblock copolymers and their corresponding strain–stress curves. The

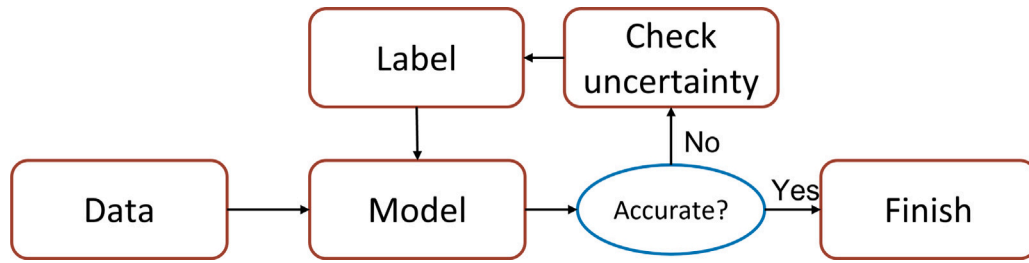


Fig. 5. The workflow of active learning. The process is conducted iteratively, which begins with an initial model, followed by the measurement of uncertainty for all data points. The most uncertain data point is then selected and labeled. Subsequently, the model is updated with this new information, and the process is repeated. This iterative cycle continues to enhance the model's performance.

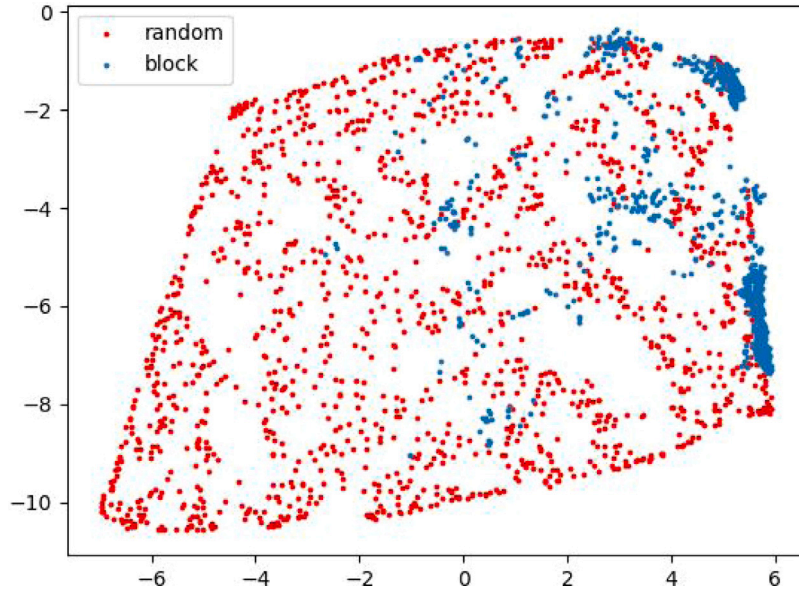


Fig. 6. The data distribution of copolymers. Random copolymers are represented as red dots, while block copolymers are represented as blue dots.

sampling strategy for random copolymers is detailed in Section 2.2, resulting in 1550 unlabeled random sequences within the dataset.

The parameters to describe a random copolymer are described in Section 2.4.1. Due to the seven-dimensional space as the surrogate model input of the dataset, understanding its distribution directly proves challenging. To address this, we employed Uniform Manifold Approximation and Projection for Dimension Reduction (UMAP) [45, 46], a general dimension reduction technique, facilitating the visualization of the copolymer's dataset. The dimensionality was reduced to two dimensions, as illustrated in Fig. 6. The results reveal that random copolymers disperse across the entire reduction space. In contrast, block copolymers predominantly occupy the right side.

As anticipated, the random copolymers exhibit a broader distribution than block copolymers despite only having labeled data for the latter. Due to their distribution, this poses challenges in constructing a surrogate model for random copolymers. The primary objective of this research is to predict the stress-strain curve of random copolymers. Consequently, we aim to transfer information from labeled block copolymers to unlabeled random copolymers.

3.2. Results of active learning

Before the active learning process, we needed to have a pre-trained supervised learning model trained on parts of data. In this study, we used block copolymer data to train an initial pre-trained model. The first step of this task was to turn the block sequence to the seven-dimension model input and then train a GPR model. All of the 1200 ABA-triblock copolymer data were used in this step. Then, we obtained

a GPR model based on the knowledge of the block copolymer input and their corresponding stress-strain curve.

The second step was to transfer information from labeled copolymers to unlabeled ones. In this step, we employed uncertainty-based active learning to efficiently sample data instead of random sampling. The active learning process was carried out iteratively. Once we had initially trained a Gaussian Process Regression model on the block copolymers model, we tested all the random copolymers data to estimate uncertainty. Then, we selected data with the highest uncertainty, represented by the standard error outputted by the GPR model, and labeled that data.

The labeling method involved CGMD, with details of the simulation method provided in Section 2.3. Following the simulation, we obtained the stress-strain curve as the previously most uncertain data label. Subsequently, we taught the model with this simulation result and completed one iteration. Afterward, we queried for the next data with the highest uncertainty for sampling and repeated the whole process.

This iterative process allowed the surrogate model to improve with each iteration. We examined the mean square error (MSE) between the model prediction and the simulation result to gauge the improvement. The MSE is defined by:

$$MSE = \sum_{i=0}^n (\sigma_{ML}(i) - \sigma_{MD}(i))^2 \quad (5)$$

where σ_{ML} is the stress predicted by the surrogate model, and σ_{MD} is the stress predicted by simulation. n starts from 0 to 101, which indicates strain from 0 to 3. Additionally, monitoring uncertainty at each step visually represents the model's enhancement. In this research, both

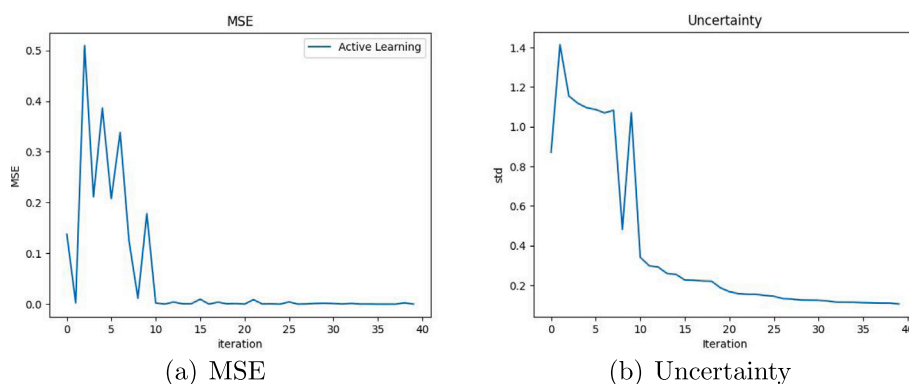


Fig. 7. (a) Mean Square Error between model prediction and simulation result. (b) Uncertainty in the process of active learning. Both of the figures show that the model converges after 40 iterations.

methods were employed to estimate active learning results, presented in Fig. 7.

The results indicate that the disparity between model predictions and simulation results becomes minimal after ten iterations and exhibits little change throughout the 30 iterations. Additionally, uncertainty diminishes substantially after 30 iterations, showing minimal changes in the subsequent ten iterations. These findings suggest that the active learning process converges after 40 iterations.

3.3. Data selected by active learning

Following the active learning process, we proceeded to visualize how uncertainty-based active learning selects data for labeling. The statistics of the data selected through the active learning process and the design space are shown in Fig. 8. It can be observed that for length, most of the selected data lies at both ends. However, there are no specific rules for active learning for the other six parameters. We employed the same approach to address this issue, utilizing UMAP for dimension reduction. The selected data was then plotted in the data distribution figure., shown in Fig. 9.

Observations from the figure lead to the conclusion that most selected data is positioned at the boundary of the space. Essentially, uncertainty-based active learning transforms the task into an interpolation problem. Despite the seven-dimensional nature of the input data shown in Table 1, which is challenging to visualize directly, the uncertainty strategy effectively identifies data with the highest value for labeling. This capability of active learning proves particularly beneficial when confronted with a combination explosion problem or when labeling data is prohibitively expensive.

In this study, the active learning strategy allows for labeling only 40 data points within the design space of 1550 data points. This efficient selection process significantly reduces the labeling cost, demonstrating the utility of active learning in scenarios where the labeling process is resource-intensive or impractical for the entire dataset.

3.4. Inference

We discussed how we uniformly sample data from all random copolymer sequences. Furthermore, we successfully constructed the surrogate model using this sampled dataset through uncertainty-based active learning. The predicted result of the training data via the surrogate model is shown in Fig. 10. We tested other sequences not included in the training set to assess the model's predictability beyond the sampled sequences.

Randomly selecting multiple data points that the surrogate model did not encounter, we allowed the trained model to predict their stress-strain curves directly. The chosen data have 60, 90, 120, 50, and 130 lengths, respectively. The components of the five samples are listed in

Table 2

Component of five samples selected to test.

	Sample1	Sample2	Sample3	Sample4	Sample5
Number of A	45	45	30	30	70
Number of B	15	45	90	20	60

Table 2, and the sequences of the five samples are illustrated in Fig. 11. The first three samples were intended to assess whether the model can accurately predict data within the length range of the training dataset. The last two samples aimed to test the model's capability to predict data beyond the range covered by the training dataset.

Subsequently, we formatted the sequences appropriately and inputted them into the trained surrogate model while concurrently running CGMD simulations. The stress-strain curves obtained from both methods were plotted simultaneously, revealing a high level of consistency between the two approaches. The result is shown in Fig. 12. Our results demonstrate that the surrogate model effectively predicts simulation outcomes across various regions, encompassing the elastic region, yielding point, and stress-hardening region. This consistency holds for polymer lengths within the specified design space and those slightly outside the designated range.

The findings conclude that we have successfully developed a surrogate model capable of accurately predicting the mechanical properties of all random copolymer sequences within the design space. We can achieve property prediction and sequence design for random copolymers using this surrogate model.

4. Conclusions

We have successfully established a standardized data science process for predicting random copolymer properties. This comprehensive process encompasses sampling strategy, active learning, surrogate modeling, and simulation. One of the primary challenges in applying data science methods to molecular systems lay in the vast design space and limited labeled data availability. Whether the data is derived from traditional experiments or computational simulations, constructing the dataset requires strategic sampling methods.

We employed specific sampling strategies to navigate the expansive random copolymer sequence universe, forming an unlabeled dataset. Instead of resorting to random sampling or exhaustive labeling, we adopted an active learning approach, considering the model's uncertainty feedback to select which data to label next. This strategic approach allowed us to construct our surrogate model in 40 iterations, utilizing a dataset comprising 1550 unlabeled random copolymer entries. After completing the entire process, we employed UMAP to analyze the efficiency of uncertainty-based active learning in sampling. Upon examining the UMAP plot, we observed that the uncertainty-based active learning method automatically sought data points on the

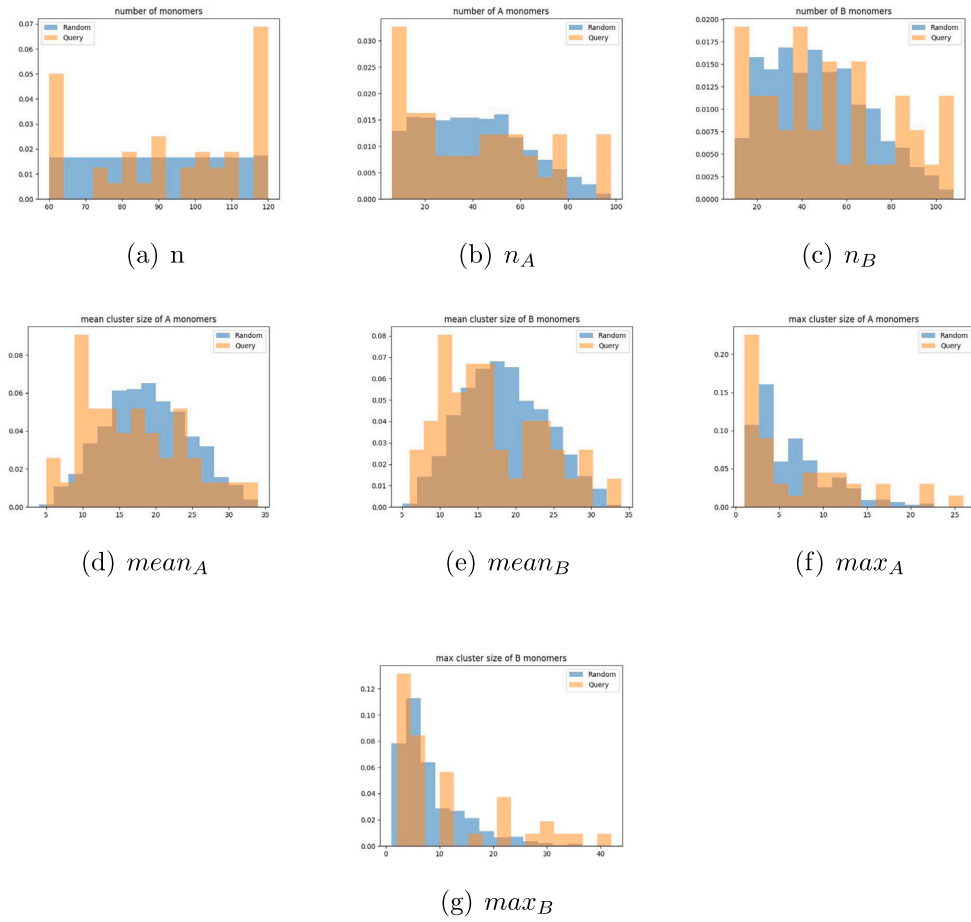


Fig. 8. The distribution of query data for seven parameters.

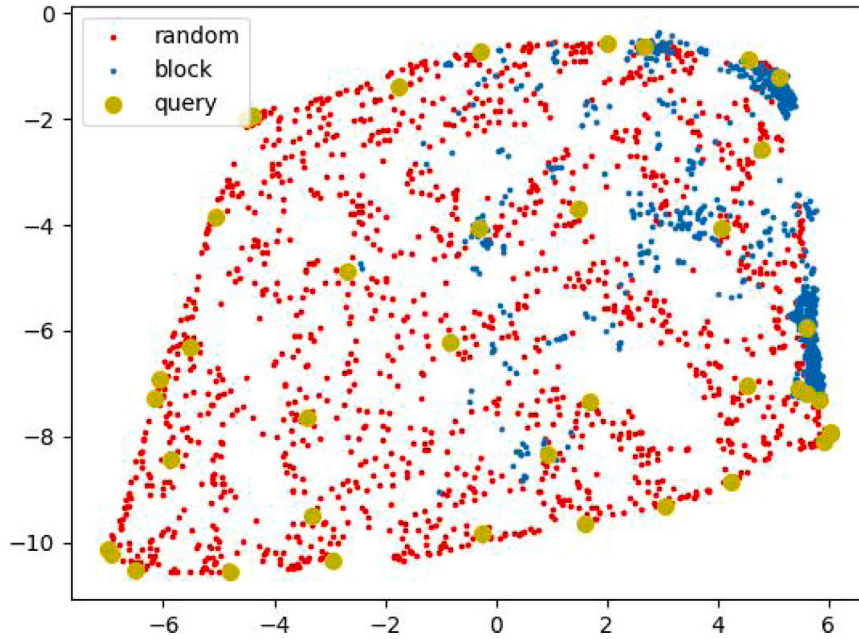


Fig. 9. Visualization of data selected by active learning. Most selected data is positioned at the boundary of the space.

design space's periphery. This strategic querying aided in transforming the extrapolation challenge into an interpolation task, subsequently enhancing the predictability of the surrogate model.

Furthermore, we evaluated the predictive capabilities of our model on unsampled data slightly beyond the design space range. The results demonstrated that the model trained through uncertainty-based active

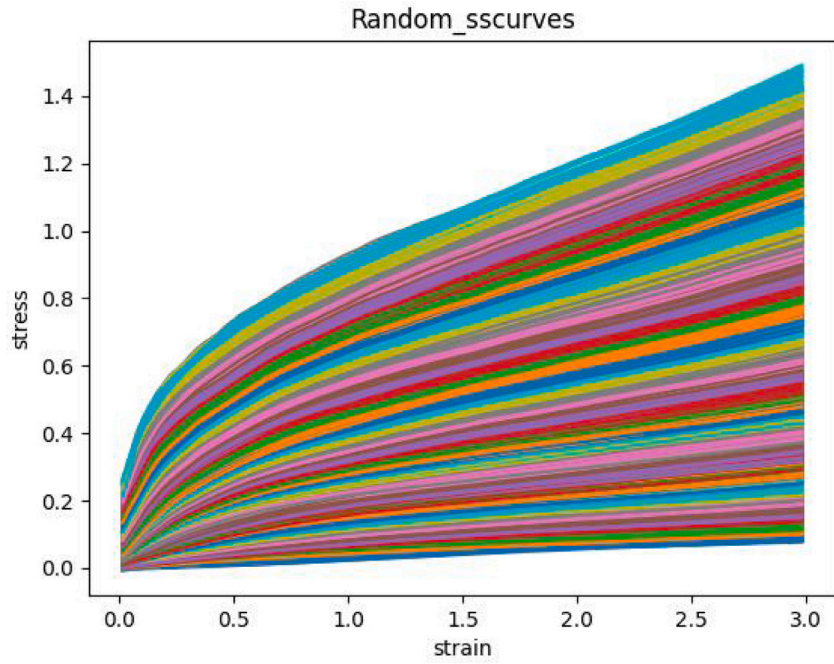


Fig. 10. The prediction result for all the data in the dataset from GPR after 40 active learning iterations.

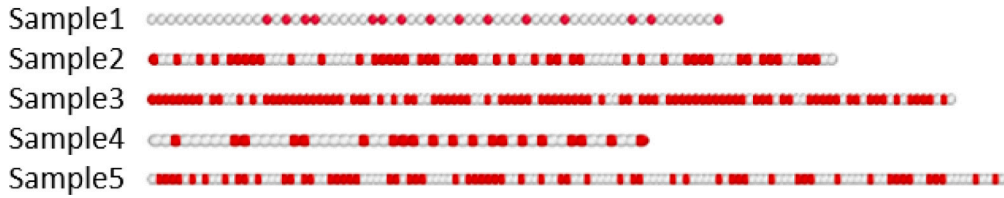


Fig. 11. The sequences of the five samples. The white beads represent the A monomers, and the red beads represent the B monomers.

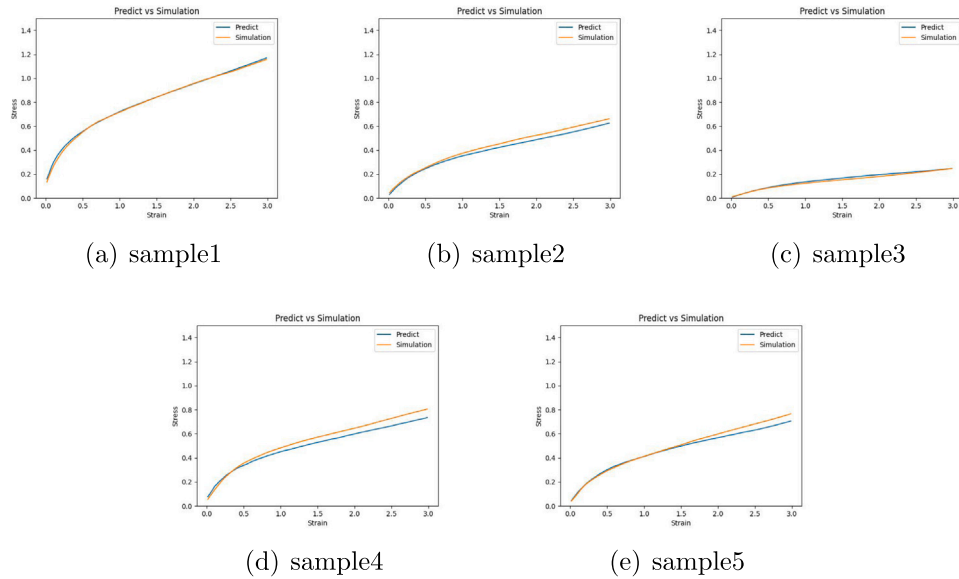


Fig. 12. The stress-strain curve predicted by surrogate model and CGMD simulation.

learning could effectively predict all the randomly selected data. This success underscores the efficiency and accuracy achieved by our data science methodology in predicting random copolymer properties.

This method holds great potential for application to other polymeric materials and for predicting and optimizing various properties,

especially in scenarios where labeling all data points within our target design space proves challenging. Although this research focuses on simulation data, we believe the same approach can be applied to guide experimental design. For example, Kraisornkachit et al. [47] demonstrated that active learning based on entirely experimental data

can efficiently guide experiments to find the optimal epoxy. Since labeling experimental data is more costly than labeling simulation data, initial knowledge can be gathered from the literature, and a limited number of experiments can be conducted to guide the learning process of the surrogate model. In our case, the process converged after approximately 40 iterations within a design space of 1550, a manageable number of experiments. After completing active learning, the surrogate model can be used for fast screening or inverse design of the polymers. Methods such as Bayesian optimization [15], genetic algorithms [20], reinforcement learning [48], and generative models [18] have been used for material optimization. Therefore, applying this framework can effectively support experiment-driven copolymer sequence design or other polymer design scenarios.

CRediT authorship contribution statement

Wei-Che Chang: Writing – original draft, Methodology, Formal analysis, Data curation. **Zong-Yun Tsai:** Investigation, Data curation. **Chin-Wen Chen:** Writing – review & editing, Supervision, Investigation, Conceptualization. **Chi-Hua Yu:** Writing – review & editing, Supervision, Investigation. **Chuin-Shan Chen:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The codes and data are deposited in Github: https://github.com/CMMAI/Mechanical_Properties_Prediction_of_Random_Copolymers.

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